

HESOD: HIGH-RESOLUTION EFFICIENT SMALL OBJECT DETECTION VIA SEMANTIC-SPECTRAL DUAL SELECTION

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ABSTRACT

Detecting tiny objects in high-resolution aerial and maritime imagery is severely challenged by background redundancy and extreme scale variation. While spatial selective computation routes foreground regions to cut latency, existing single-evidence selectors suffer from severe semantic collapse on micro targets, causing irrecoverable coarse-stage pruning. We propose **HESOD** (High-Resolution Efficient Small Object Detection), a dual-evidence framework featuring semantic-spectral routing, coverage-constrained spatial optimization, and lightweight downstream detection. HESOD couples semantic objectness with constant-width spectral saliency under a monotonic evidence-preserving max-fusion rule to rescue degraded targets. To promote recall-safe routing, an object-level soft-coverage loss directly optimizes multi-cell joint instance coverage across ground-truth objects. To recover downstream efficiency, an inverted residual decoupled head (ISPP) is integrated to compress parameters by $> 27\%$ with only fractional FLOPs overhead. Extensive experiments show that HESOD improves mAP@.5 by 0.9 pp and 2.3 pp (boosting total recall by 3.66 pp and 3.33 pp) on UAVDT and SeaPerson, respectively, establishing an effective accuracy-efficiency co-design suitable for resource-constrained deployment.

Index Terms— Small object detection, selective computation, spectral saliency, coverage optimization, UAV

1. INTRODUCTION

High-resolution aerial and maritime vision is fundamental to intelligent surveillance, environmental monitoring, and search-and-rescue. In operational scenarios across aerial and maritime domains [1–3], targets of interest typically occupy only a handful of pixels ($< 16 \times 16$ px) amid massive uniform backgrounds ($> 70\%$ – 80%) [4, 5]. Evaluating standard dense deep detectors across multi-megapixel grids (1280×1280 to 2048×2048) incurs prohibitive arithmetic latency on resource-constrained platforms.

To eliminate redundant computation, *spatial selective computation* dynamically routes feature processing strictly to prospective foreground regions [4, 6, 7]. Notably, ESOD [4] predicts an early spatial priority heatmap directly after a shallow stem, slicing features into candidate bounding patches (AdaSlicer) and bypassing background across downstream stages.

Crucially, spatial selection operates under a severe *asymmetric error cost*: false-positive background patches merely incur minor compute overhead, whereas discarding a true tiny target produces an *irrecoverable false negative* across downstream stages [4]. Existing detectors rely exclusively on single semantic objectness with independent per-cell BCE. Under extreme downsampling, micro targets suffer severe semantic collapse, causing faint instances to be prematurely pruned (Fig. 1(b)). To overcome this recall dilemma, we

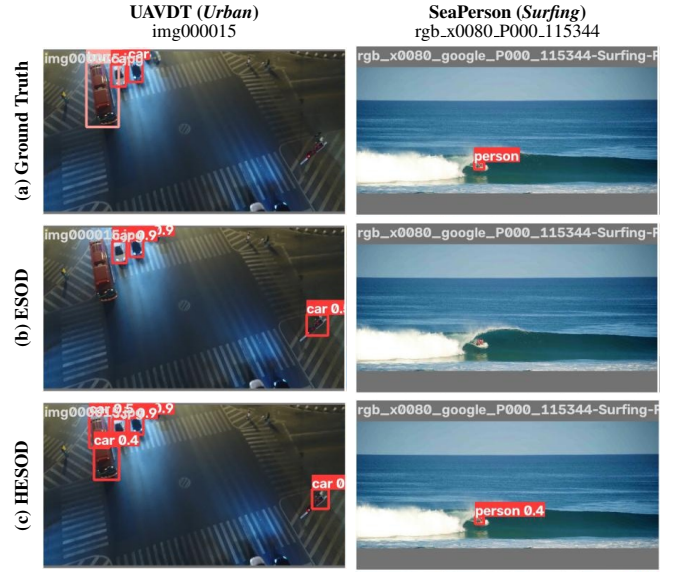


Fig. 1. Visual comparison on UAVDT and SeaPerson. Baseline ESOD drops micro targets (b), while HESOD restores full instance recall (c).

observe that tiny targets retain prominent high-frequency spatial gradients and spectral saliency even when deep semantics fade [8–10]. We reveal that naive selective inference suffers from acute semantic collapse under asymmetric pruning risks, and propose **HESOD** (High-Resolution Efficient Small Object Detection) to establish recall-safe, evidence-preserving sparse routing. Our key contributions are:

1. **Dual-Evidence Spatial Routing:** We uncover single-semantic routers’ vulnerability to feature collapse, coupling semantic objectness with constant-width gradient filtering for scale-robust structural priors independent of backbone capacity.
2. **Evidence Preservation and Soft Coverage:** We formulate a monotonic max-fusion rule preventing multi-cue evidence dilution, paired with an object-level noisy-OR soft-coverage loss directly optimizing joint instance recall under irreversible pruning.
3. **Architectural Compute Compensation:** We co-design downstream detection with an inverted residual decoupled head [11] as a dedicated compute compensator, amortizing candidate evaluation overhead while slashing detector parameters by $> 27\%$.

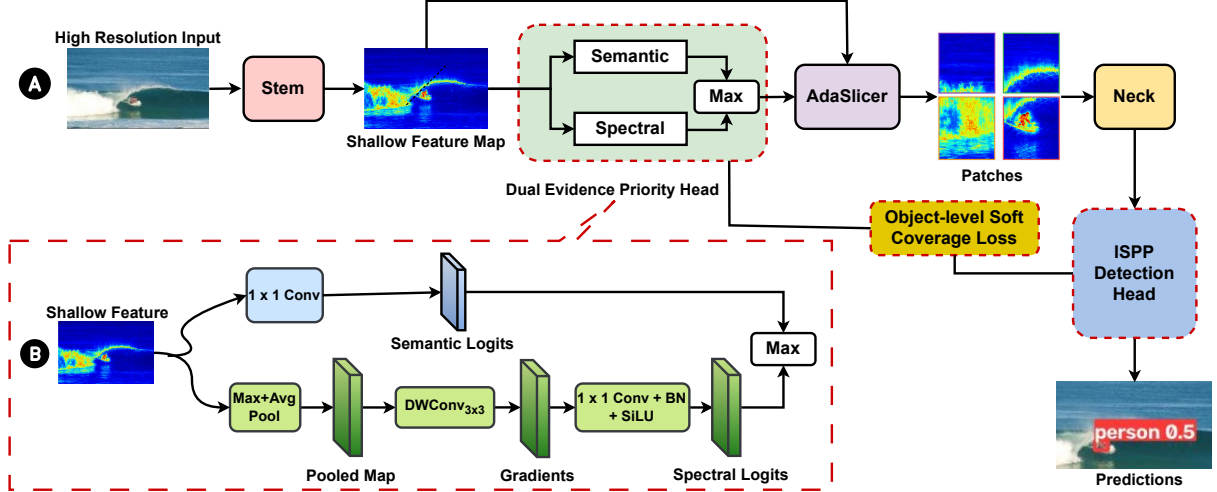


Fig. 2. Overview of HESOD: (a) End-to-end selective pipeline; (b) Dual-Evidence Priority Head. Dashed boxes highlight our proposed contributions over ESOD.

2. RELATED WORK

2.1. Small Object Detection in High-Resolution Vision

Detecting tiny targets in high-resolution aerial and maritime imagery is severely challenged by extreme scale disparity, low contrast, and massive background areas ($> 70\%$ – 80%) [1–3, 12]. Classical approaches utilize density-guided cropping (ClusDet [13], DMNet [14]) or scale-adaptive label assignment (NWD [15], RFLA [16]). Recently, ResBiDet [17] introduced a coarse-to-fine framework using global heatmap guidance for region cropping. However, it relies on multi-scale image-level slicing and boundary stitching. In contrast, HESOD performs early feature-level selective routing directly within the detector trunk and explicitly optimizes instance coverage under irreversible pruning.

2.2. Selective and Dynamic Computation

Dynamic computation accelerates inference by allocating compute strictly to prospective foreground regions (e.g., AutoFocus [5], QueryDet [6], CEASC [7], and sparse location activation [18]). Recently, coarse-to-fine detectors such as ESOD [4] and OSDT [19] predict early spatial priority heatmaps to extract candidate patches. However, existing selectors rely exclusively on a single semantic score supervised by independent per-cell BCE. Under extreme scale disparity, micro targets suffer from semantic feature collapse after downsampling; meanwhile, per-cell BCE lacks joint instance-level coverage constraints, causing fatal false negatives under an asymmetric error cost. HESOD resolves this by combining semantic and spectral dual cues with object-level soft-coverage optimization.

2.3. Spectral Saliency and Frequency Methods

Frequency-domain representations offer rich structural priors for micro targets, which retain sharp high-frequency edge gradients even when photometric semantics fade [8–10]. While recent methods explore dense frequency transforms [8] or detail compensation across multi-scale feature pyramids [20], they incur substantial downstream latency. In sharp contrast, HESOD repurposes high-frequency gradient saliency upstream as an ultra-lightweight routing cue: channel pooling compresses stem features to 2 channels, enabling directional

Laplacian and Sobel gradient filtering at constant width, independent of backbone channel dimension C .

3. PROPOSED METHOD

The architecture of HESOD is shown in Fig. 2(a). From input $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, a shallow stem extracts dense features $\mathbf{F} \in \mathbb{R}^{H_s \times W_s \times C}$. The *Dual-Evidence Spatial Selector* then coordinates: (i) a priority head fusing semantic objectness with channel-pooled spectral saliency (Fig. 2(b)) into spatial priority map $\mathbf{S} \in [0, 1]^{H_s \times W_s}$; and (ii) an AdaSlicer module dynamically clustering candidate foreground cells into bounding patches $\mathcal{P}$ from $\mathbf{F}$. Downstream backbone stages and lightweight decoupled heads evaluate only retained patches $\mathcal{P}$ under soft-coverage supervision (§3.2, §3.3).

3.1. Dual-Evidence Priority Estimation

To rescue micro targets with degraded semantic activations, HESOD couples high-level semantic confidence with low-level structural gradient saliency at negligible compute overhead (Fig. 2(b)).

Let $\mathbf{f}_i \in \mathbb{R}^C$ denote the feature vector at spatial location i in $\mathbf{F}$. The semantic branch predicts class-specific logits via a 1×1 convolution: $\mathbf{z}_i^{\text{sem}} = \text{Conv}_{1 \times 1}(\mathbf{f}_i) \in \mathbb{R}^{N_c}$. In parallel, the spectral branch extracts high-frequency structural gradients by first compressing channels via max- and mean-pooling:

$$\mathbf{F}_{\text{pooled}} = \left[\max_c \mathbf{F}_{:, :, c} \parallel \frac{1}{C} \sum_{c=1}^C \mathbf{F}_{:, :, c} \right] \in \mathbb{R}^{H_s \times W_s \times 2}, \quad (1)$$

where $[\cdot \parallel \cdot]$ denotes channel concatenation. Compressing $\mathbf{F}$ to 2 channels retains peak edge activation and background energy with $\mathcal{O}(H_s W_s)$ complexity independent of width C . Next, a learnable depthwise 3×3 convolution bank—initialized with discrete Laplacian and directional Sobel operators—filters $\mathbf{F}_{\text{pooled}}$ into 6 structural channels. In the 2D spatial frequency domain, the discrete Laplacian acts as an isotropic high-pass filter with $|H(\omega)| \propto \|\omega\|^2$, naturally amplifying edge transitions. Because micro targets ($< 16^2$ px) manifest as localized high-frequency spatial impulses whose gradient energy resists downsampling [10], this branch supplies a constant-bandwidth prior independent of deep semantic capacity. A 1×1 projection maps features to N_c , yielding spectral logits $\mathbf{z}_i^{\text{spec}} \in \mathbb{R}^{N_c}$.

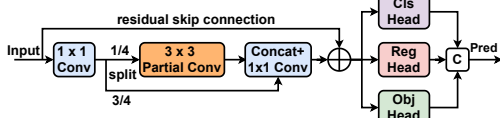


Fig. 3. Inverted Residual Decoupled Head (ISPP) with Partial Convolution (© denotes channel concatenation).

To prevent confidence dilution on faint targets, evidence fusion enforces *evidence preservation*. Linear combinations risk destructive interference: an unconfident or negative activation in one branch can drag down a confident signal in the other below the routing threshold. Instead, we formulate a parameter-free elementwise maximum:

$$z_{i,c} = \max(z_{i,c}^{\text{sem}}, z_{i,c}^{\text{spec}}), \quad s_i = \max_{c=1}^{N_c} \sigma(z_{i,c}), \quad (2)$$

This operation functions as an evidence-preserving OR gate: since $z_{i,c} \geq z_{i,c}^{\text{sem}}$ and $z_{i,c} \geq z_{i,c}^{\text{spec}}$, neither branch can ever suppress genuine candidate cues. Cells with priority $s_i > \tau$ are clustered into bounding patches $\mathcal{P}$ for downstream detection.

3.2. Object-Level Soft-Coverage Loss

Standard pixel-wise binary cross-entropy (BCE) evaluates grid cells independently, failing to guarantee instance-level coverage: a selector may over-allocate cells to large objects while dropping micro objects whose receptive fields contain only weak responses. To prevent premature false negatives, we formulate a differentiable object-level soft-coverage loss.

Let $\mathcal{N}(j)$ denote the set of selector cells overlapping ground-truth box $j \in \{1, \dots, N_{\text{gt}}\}$. We formulate a noisy-OR differentiable coverage surrogate and its weighted instance loss:

$$p_j^{\text{cover}} = 1 - \prod_{i \in \mathcal{N}(j)} (1 - s_i), \quad \ell_j = -w_j \log(p_j^{\text{cover}} + \epsilon), \quad (3)$$

where $w_j = \text{clip}(4/a_j, 1, 5)$ is a scale-adaptive weight based on bounding box area a_j in cells, penalizing misses on micro targets ($a_j < 4$). For any cell $i \in \mathcal{N}(j)$, the gradient of ℓ_j with respect to cell score s_i is:

$$\frac{\partial \ell_j}{\partial s_i} = -\frac{w_j}{p_j^{\text{cover}} + \epsilon} \prod_{k \in \mathcal{N}(j) \setminus \{i\}} (1 - s_k). \quad (4)$$

When neighboring candidate cells have low scores ($s_k \rightarrow 0$), the product term approaches 1, generating a strong gradient to boost cell i and encouraging at least one cell to activate per instance. The overall soft-coverage loss averages over instances:

$\mathcal{L}_{\text{cover}} = \frac{1}{N_{\text{gt}}} \sum_{j=1}^{N_{\text{gt}}} \ell_j$. The multi-task training objective is $\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_{\text{sel}}(\mathcal{L}_{\text{BCE}} + \lambda_{\text{cov}}\mathcal{L}_{\text{cover}})$, with $\lambda_{\text{sel}} = 0.2$ and $\lambda_{\text{cov}} = 0.5$.

3.3. Lightweight Decoupled Head and Training Protocol

While dual-evidence routing substantially improves patch coverage by retaining faint foreground regions, evaluating more candidate patches increases downstream detector computation. To recover this overhead without sacrificing selector gains, HESOD incorporates an inverted residual partial-convolution (ISPP) decoupled head [11] as a dedicated compute compensator.

As shown in Fig. 3, the ISPP head expands neck feature channels to $2C_1$, partitioning them into an active branch (0.25 ratio) processed by a 3×3 Partial Convolution (PConv) and an identity passthrough branch (0.75 ratio). After linear projection and residual addition, decoupled 1×1 heads predict classification, box regression, and objectness independently, lowering arithmetic complexity while eliminating inter-task gradient interference.

Two-Stage Training Protocol. To ensure stable routing behavior, HESOD employs a two-stage training scheme: (i) *Stage 1 (Selector Convergence)*: Train the Dual-Max selector with a coupled head under $\mathcal{L}_{\text{cover}}$, establishing a well-calibrated priority heatmap; and (ii) *Stage 2 (Head Fine-Tuning)*: Freeze the upstream trunk (backbone, dual branches, router) and fine-tune the lightweight ISPP neck and decoupled heads on the frozen routing distribution. Relative to the uncompressed Dual-Max selector, this lightweight head slashes GFLOPs by 16.9% and parameters by 27.5% (on UAVDT) while fully preserving peak coverage (0.940 BPR), maintaining competitive real-time FPS with only fractional overhead vs. ESOD.

4. EXPERIMENTS

4.1. Experimental Design and Evaluation Criteria

We evaluate HESOD on two demanding high-resolution small-object benchmarks: (1) UAVDT [2]: 1280×1280 px urban aerial imagery for vehicle surveillance across 3 classes, where extremely tiny targets ($< 16^2$ px) account for 19.9% of all instances; (2) SeaPerson [3, 12]: 2048×2048 px maritime search-and-rescue imagery with over 85% of targets smaller than 32 px, accompanied by large-area sea surface background redundancy and low-contrast interference. Both benchmarks use full-resolution images without artificial cropping, fully simulating real edge deployment scenarios.

Selective detectors (ESOD family and our ablations) employ YOLOv5m backbones in PyTorch 2.8 trained for 50 epochs under selective protocols [4]. The proposed two-stage training scheme is implemented as follows: (i) Stage 1 (Selector Convergence): Train the Dual-Max selector with a coupled head under soft coverage loss, establishing a well-calibrated priority heatmap; (ii) Stage 2 (Head Fine-Tuning): Freeze the upstream trunk (backbone, dual branches, router) and fine-tune the lightweight ISPP neck and decoupled heads on the frozen routing distribution to ensure stable routing behavior. Dense and query baselines (Faster R-CNN, RetinaNet, QueryDet) retain canonical ResNet-50-FPN backbones, evaluated on identical test splits and resolutions. All models are benchmarked on an NVIDIA RTX 5090 GPU. Metrics include: (1) Standard COCO metrics: mAP@.5, mAP@.5:.95; (2) Scale-stratified recall metrics: recall is counted across four bins: Very Tiny ($< 16^2$ px), Tiny ($16^2 - 32^2$ px), Small ($32^2 - 96^2$ px), and Medium/Large ($> 96^2$ px); (3) Routing-specific metric: Bounding Patch Recall (BPR), defined as the proportion of ground-truth targets whose overlapping area with selected patches exceeds 50% of their own area, which directly measures the missed detection risk of the selector.

4.2. Main Benchmark Comparisons

4.2.1. Comparison with Generic Detection Baselines

As shown in Table 1, traditional dense detectors suffer from severe efficiency bottlenecks on high-resolution images: Faster R-CNN only achieves 18.8 FPS and 23.1 FPS on UAVDT and SeaPerson respectively, with GFLOPs as high as 772.5 and 1546.8, which cannot meet real-time deployment requirements on edge devices; sparse detection methods such as QueryDet [6] are limited by the dynamic query mechanism, and their recall drops significantly in small object dense scenarios, with total recall only 41.89% on UAVDT.

Our HESOD achieves comprehensive breakthroughs in both accuracy and efficiency compared with all generic baselines: On UAVDT, mAP@.5 is 5.8 pp, 9.8 pp, and 13.7 pp higher than Faster R-CNN, RetinaNet, and QueryDet respectively, and the inference

Table 1. Benchmark results on UAVDT and SeaPerson.

Method	mAP@.5	Params (M)	GFLOPs	Total Recall	BPR	FPS
UAVDT Benchmark (1280 × 1280)						
Faster R-CNN [21]	0.336	43.3	772.5	79.68%	–	18.8
RetinaNet [22]	0.296	36.4	360.9	83.44%	–	19.8
QueryDet [†] [6]	0.257	39.3	– [†]	41.89%	–	21.1
ESOD [4]	0.385	35.9	68.2	85.17%	0.884	117.8
HESOD (Ours)	0.394	26.0	<u>74.9</u>	88.83%	0.940	<u>106.1</u>
Δ vs. ESOD	+0.9 pp	-27.5%	+9.8%	+3.66 pp	+5.6 pp	-9.9%
SeaPerson Benchmark (2048 × 2048)						
Faster R-CNN [21]	0.551	43.3	1546.8	67.27%	–	23.1
RetinaNet [22]	0.473	36.4	942.7	65.98%	–	28.0
QueryDet [†] [6]	0.627	39.3	– [†]	61.56%	–	14.0
ESOD [4]	0.750	35.8	202.4	84.42%	0.947	85.7
HESOD (Ours)	0.773	25.9	<u>208.5</u>	87.75%	0.991	<u>80.6</u>
Δ vs. ESOD	+2.3 pp	-27.6%	+3.0%	+3.33 pp	+4.4 pp	-6.0%

BPR: Bounding Patch Recall. **Bold**: best; underline: second-best (selective models).

[†]QueryDet uses CSQ (dynamic sparse FLOPs unamenable to static profiling; dense query counterparts: 973.5 / 2539.0 GFLOPs at 12.5 / 10.6 FPS on UAVDT / SeaPerson).

speed reaches 106.1 FPS, which is $5.0\times\text{--}5.6\times$ higher than the baselines; On SeaPerson, mAP@.5 exceeds the baselines by 14.6~30.0 pp, FPS reaches 80.6, which is $2.9\times\text{--}5.8\times$ higher than the baselines, among which the recall of Very Tiny targets ($< 16^2$ px) reaches 75.71%, 27.0 pp higher than QueryDet, verifying the recall ability of dual-evidence routing for micro targets.

4.2.2. Comparison with Selective Detection Baseline (ESOD)

Against the same-architecture selective detection baseline ESOD [4], HESOD achieves significant accuracy improvement with only a small computational overhead, while greatly compressing the parameter scale:

- UAVDT Benchmark: mAP@.5 increases from 0.385 to 0.394 (+0.9 pp), Bounding Patch Recall (BPR) increases from 0.884 to 0.940 (+5.6 pp), total recall increases by 3.66 pp, where the recall of bus class (a typical few-shot tiny target) increases from 67.7% to 81.4%, verifying the robustness of the model to few-shot extremely tiny targets; the parameter scale is reduced from 35.9M to 26.0M, a 27.5% compression, and the inference speed is maintained at 106.1 FPS, only 9.9% lower than ESOD, meeting real-time deployment requirements.
- SeaPerson Benchmark: mAP@.5 increases from 0.750 to 0.773 (+2.3 pp), BPR reaches 0.991 (+4.4 pp), total recall increases by 3.33 pp, successfully recalling more than 13,700 micro maritime targets missed by ESOD; GFLOPs only increase by 3.0%, parameters are compressed by 27.6%, and the inference speed is maintained at 80.6 FPS, only 6.0% lower than ESOD.

The slight increase in GFLOPs of HESOD comes from the selector recalling more real target regions rather than invalid computational redundancy: the low GFLOPs of ESOD is at the cost of early discarding micro targets. Under the premise of ensuring more than 99% target coverage, HESOD controls the additional computational overhead within 10% through the lightweight head design, achieving the optimal balance between “recall and efficiency”.

4.3. Ablation Studies and Scale Analysis

To quantitatively analyze the independent contribution of each module, we conduct controlled variable experiments by adding components one by one on the basis of ESOD baseline, and the results are shown in Table 2.

4.3.1. Effectiveness of Dual-Evidence Fusion Mechanism

When using only the semantic branch, the BPR on UAVDT is 0.906, and the total recall is 84.82%, which has limited improvement compared with the baseline, because the semantic features of small targets collapse during downsampling; when using only the spectral branch, the BPR increases to 0.942, and the total recall reaches 88.87%, verifying the robustness of spectral gradient features to micro targets.

Comparing the two fusion strategies: the linear fusion method using channel concatenation only achieves 0.371 mAP@.5, which is even lower than the single-branch effect, because weak semantic responses will weaken the strong activation signals of the spectral branch, resulting in “evidence dilution”; the proposed element-wise maximum fusion (Dual-Max) implements an evidence-preserving OR gate effect, achieving 0.395 mAP@.5 and 0.940 BPR on UAVDT, 0.778 mAP@.5 and 0.991 BPR on SeaPerson, which is the optimal value among all single-stage configurations, verifying the effectiveness of the evidence-preserving fusion principle.

4.3.2. Effectiveness of Object-Level Soft-Coverage Loss

When the ESOD baseline uses per-pixel BCE loss to train the selector, the BPR on UAVDT is only 0.884, and the recall of Very Tiny targets is only 79.43%, with severe micro target missed detection. After replacing it with the proposed soft-coverage loss, the Very Tiny recall of the semantic branch increases to 79.13%, and that of the spectral branch increases to 82.43%. Under the Dual-Max fusion configuration, the Very Tiny recall reaches 85.69%, 6.26 pp higher than the baseline. Analysis of the loss gradient shows that when the activation scores of all cells covering a target are low, the soft-coverage loss will generate a strong gradient signal to boost the activation value of at least one cell, ensuring that each target is covered by at least one candidate region from the training mechanism, fundamentally reducing the risk of unrecoverable missed detection.

4.3.3. Efficiency Compensation Effect of Lightweight ISPP Head

The Dual-Max configuration without the ISPP head achieves 90.1 GFLOPs and only 102.3 FPS on UAVDT. After adding the ISPP head, GFLOPs decrease by 16.9% to 74.9, FPS increases to 106.1, and the parameter scale is compressed by 27.5%, while mAP@.5 only decreases by 0.001, and BPR remains completely unchanged at 0.940; on SeaPerson, GFLOPs decrease by 18.2%, parameters are compressed by 27.6%, FPS increases from 73.8 to 80.6, and mAP@.5 only decreases by 0.005, verifying that the lightweight head fully offsets the computational overhead of additional candidate regions without losing accuracy, achieving collaborative optimization of accuracy and efficiency.

4.3.4. Engineering Optimization Value of Channel Pooling

Without channel pooling, the spectral branch needs to process features with the same dimension as the backbone, the batch size can only be set to 2 on SeaPerson, the training is unstable, and mAP@.5 is only 0.763; after adding channel pooling to compress features to 2 channels, the batch size can be increased to 8, the training stability is greatly enhanced, mAP@.5 increases by 1.5 pp to 0.778, GFLOPs decrease by 8.9%, and the Very Tiny recall further increases by 0.9 pp, proving that this operation jointly improves training efficiency, inference efficiency and detection accuracy.

Table 2. Module ablation, efficiency, and scale recall breakdown on UAVDT and SeaPerson.

Configuration	Detection Accuracy and Computational Efficiency						Physical-Scale Recall Breakdown				
	mAP @.5	mAP @.5:.95	BPR	Params (M)	GFLOPs	FPS	Very Tiny ($< 16^2$)	Tiny (16^2-32^2)	Small (32^2-96^2)	Med/Lrg ($> 96^2$)	Total Recall
<i>UAVDT Benchmark</i> (1280×1280)							(74,375)	(206,015)	(86,282)	(7,325)	(373,997)
(1) ESOD Baseline*	0.385	0.214	0.884	35.9	68.2	117.8	79.43%	84.21%	94.54%	59.78%	85.17%
(2) Semantic-only	0.384	<u>0.217</u>	0.906	35.9	75.0	<u>113.8</u>	79.13%	83.91%	94.00%	59.95%	84.82%
(3) Spectral-only	0.387	0.209	<u>0.942</u>	36.0	93.7	102.3	82.43%	88.72%	96.66%	<u>66.80%</u>	88.87%
(4) (3) + Channel-Pooled	0.394	0.209	0.963	35.9	99.3	97.6	86.88%	92.23%	97.74%	66.43%	91.93%
(5) Dual-Concat	0.371	0.205	0.919	35.9	83.9	107.2	79.09%	85.85%	95.85%	62.66%	86.35%
(6) Dual-Max	0.395	0.218	0.940	35.9	90.1	102.3	<u>85.69%</u>	<u>89.97%</u>	<u>97.37%</u>	65.99%	<u>90.36%</u>
(7) HESOD (Ours)	<u>0.394</u>	0.215	0.940	26.0	<u>74.9</u>	106.1	80.76%	88.98%	97.11%	69.16%	88.83%
<i>SeaPerson Benchmark</i> (2048×2048)							(82,417)	(174,816)	(42,987)	(155)	(300,375)
(1) ESOD Baseline*	0.750	0.320	0.947	35.8	202.4	85.7	74.03%	86.78%	94.74%	79.35%	84.42%
(2) Semantic-only	0.769	0.325	0.991	35.8	266.8	73.5	75.49%	91.57%	95.71%	81.94%	87.75%
(3) Spectral-only	0.770	<u>0.327</u>	0.992	35.8	267.4	71.7	75.23%	<u>91.69%</u>	95.89%	<u>86.45%</u>	87.77%
(4) (3) + Channel-Pooled	0.767	0.324	0.988	35.8	263.4	72.7	76.13%	91.36%	95.50%	87.10%	87.77%
(5) Dual-Concat	0.772	0.326	0.986	35.8	281.2	69.8	<u>76.00%</u>	91.50%	95.68%	80.00%	<u>87.84%</u>
(6) Dual-Max	0.778	0.330	<u>0.991</u>	35.8	255.1	73.8	75.65%	92.06%	<u>95.85%</u>	81.94%	88.10%
(7) HESOD (Ours)	<u>0.773</u>	<u>0.327</u>	<u>0.991</u>	25.9	<u>208.5</u>	<u>80.6</u>	75.71%	91.47%	95.72%	80.65%	87.75%

*Trained with standard BCE (others use $\mathcal{L}_{\text{cover}}$). Parenthetical values denote ground-truth counts per bin. One-to-one matching at $\text{conf} \geq 0.001$, $\text{IoU} \geq 0.5$. **Bold**: best; underline: second-best.

5. CONCLUSION

We presented HESOD, an efficient dual-evidence framework for high-resolution small object detection under selective computation. By uniting semantic objectness with constant-width spectral saliency via monotonic max-fusion, HESOD addresses early semantic collapse on micro targets. An object-level soft-coverage loss ($\mathcal{L}_{\text{cover}}$) optimizes instance coverage under asymmetric pruning risks, while an inverted residual decoupled head (ISPP) amortizes downstream evaluation overhead. On UAVDT and SeaPerson, HESOD improves mAP@.5 by 0.9 pp and 2.3 pp (total recall +3.66 pp and +3.33 pp), cuts parameters by $> 27\%$, and achieves 106.1/80.6 FPS on an RTX 5090. Dual-evidence routing reaches 0.940/0.991 BPR, shifting the remaining bottleneck to downstream regression refinement. These results confirm that spectral structural priors safeguard coarse-stage routing, supporting practical accuracy-efficiency co-design for edge drone platforms.

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